

Genre-Locking: How Entanglement Structure Selects Gravitational Dynamics

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Different entanglement scaling laws select different gravitational dynamics. We propose the *genre-locking hypothesis*: area-law entanglement produces Einstein gravity, logarithmic (critical) entanglement produces 2D dilaton gravity, and volume-law entanglement produces no coherent semiclassical geometry. This three-way classification across scaling laws—distinct from the functional-form selection established in prior work [1, 2]—has not previously been presented as a unified phase diagram.

The framework starts from entanglement equilibrium [3] on a many-body quantum state manifold, with the area-law coefficient s_0 promoted to a dynamical field. All derived field equations are contingent on a constitutive lapse-entropy relation (Assumption A6); different constitutive choices yield different dynamics, analogous to how different equations of state yield different thermodynamic behavior. In 1+1 dimensions, this yields entanglement-dilaton gravity with s_0 as the dilaton (Appendix A). In $(d+1)$ dimensions, we propose a scalar-tensor ansatz—constrained by the 2D derivation and the vacuum Einstein limit—that reduces to vacuum Einstein gravity with cosmological constant when s_0 is constant.

The effective Newton constant is $G_{\text{eff}} = 1/(4\hbar s_0)$. Numerical confirmation on a 1D free-fermion chain ($c = 1$ recovered to within 1%, with fitting-window systematics documented) and a 2D square-lattice CDW insulator (system-size coefficient of variation $CV = 0.01\%$) demonstrates that s_0 is determined by UV physics and invariant under all IR variations to machine precision.

We make a quantitative, parameter-free prediction: the area-law coefficient near the quantum critical point scales as $s_0 = (c/6) \ln(\xi/a) + \text{const}$, yielding exact Calabrese-Cardy slopes (1/12 for the transverse-field Ising model, 1/6 for the CDW chain) and a *logarithmic* (not power-law) vanishing of the emergent curvature. This is testable by DMRG or exact diagonalization on any gapped lattice system.

I. INTRODUCTION

A. The Problem

General relativity and quantum mechanics remain unreconciled after ninety years. GR describes spacetime as a smooth, dynamical manifold curved by matter; quantum mechanics describes discrete, probabilistic states entangled across space. The emergent gravity program [3–6] proposes that this tension dissolves if spacetime and gravity are not fundamental but emergent from quantum entanglement.

B. What This Paper Adds

We construct an explicit emergent gravity framework starting from the Bures/Fubini-Study metric on a many-body quantum state manifold, and demonstrate:

1. The emergent metric’s self-consistency condition is a gravitational field equation.
2. The specific field equation depends on the entanglement scaling law—the *genre* of the underlying quantum state.

3. Area-law entanglement in $(d+1)$ dimensions produces Einstein gravity with cosmological constant in the vacuum limit.
4. The effective Newton constant is identified with the area-law coefficient s_0 , whose independence from the IR entanglement spectrum has been confirmed numerically in 1D and 2D spatial systems.
5. The 2D special case—entanglement-dilaton gravity with dilaton $\Phi = s_0$ —is recovered as a consistent limit.
6. The unified three-way scaling-law classification (area $\rightarrow$ Einstein, log $\rightarrow$ dilaton, volume $\rightarrow$ no geometry) extends and is distinct from the functional-form selection established in prior work [1, 7].

C. Relation to Prior Art

The principle that the form of the entanglement entropy selects the gravitational dynamics is well established. Faulkner *et al.* [1] showed that Ryu-Takayanagi (area) entropy yields linearized Einstein, while a general Wald functional yields higher-curvature gravity. Swingle and Van Raamsdonk [2] stated Einstein is “singled out as the unique choice given our starting assumption that the leading contribution to holographic entanglement entropy is area.” Haehl *et al.* [7] extended this to second order. Callebaut and Verlinde [8] derived 2D dilaton (JT-type) gravity from entanglement dynamics of a 2D CFT.

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Cao, Carroll, and Michalakis [9] recovered a spatial analog of Einstein’s equation from area-law “redundancy-constrained states.”

What is new here. These works establish *functional-form selection*—area-type vs. Wald functional selects Einstein vs. higher-curvature. We propose *scaling-law selection*—area vs. log vs. volume selects *entire genres* of gravitational dynamics. A Wald functional still has an area-type leading term; it is neither log-law nor volume-law. The three-way scaling-law classification as a unified phase diagram has not appeared in the published literature. The identification of the area-law coefficient s_0 as the dilaton in a self-consistency-derived field equation, supported by numerical UV-independence verification, is the primary novel contribution.

Distinction from Bianconi (2025). An alternative information-theoretic approach was recently developed by Bianconi [10], who derives modified gravitational equations from the Geometric Quantum Relative Entropy (GQRE). Our framework differs in its use of entanglement (von Neumann) entropy rather than relative entropy, in its constitutive lapse-entropy relation connecting quantum state data to Lorentzian geometry, and in the genre-locking classification that maps entanglement scaling laws to distinct gravitational theories—a structure absent from the GfE framework.

Distinction from Bianchi & Myers (2014). Bianchi and Myers [11] conjectured that area-law entanglement is the defining characteristic of quantum states with a semiclassical geometric dual. Our framework formalizes this conjecture for the area-law arm, derives the corresponding field equations, and extends the classification to log-law and volume-law regimes—directions Bianchi-Myers does not address.

D. Epistemic Framework

All claims are tiered:¹

- **Tier 1:** Verified mathematics or confirmed numerics.
- **Tier 2:** Framework-internal derivations.
- **Tier 3:** Interpretive extensions or conjectures.

II. THE CONSTRUCTION

A. The State Manifold

Consider a family of many-body ground states $|\Psi(g)\rangle$ parameterized by couplings g (e.g., the gap Δ and hop-

ping t in a lattice Hamiltonian). The Bures/Fubini-Study metric on this family defines a Riemannian geometry:

$$ds_{\text{FS}}^2 = G_{ij}(g) dg^i dg^j, \quad (1)$$

where $G_{ij} = \text{Re}[\langle \partial_i \Psi | \partial_j \Psi \rangle - \langle \partial_i \Psi | \Psi \rangle \langle \Psi | \partial_j \Psi \rangle]$ is the quantum geometric tensor.

B. The Area Law

For a gapped many-body state, the entanglement entropy of a spatial region R satisfies

$$S_{\text{ent}}(R) = s_0 \cdot A(\partial R) + \text{subleading}, \quad (2)$$

where $A(\partial R)$ is the $(d-1)$ -dimensional area of the boundary and s_0 is the area-law coefficient, playing the role of $1/(4G_{\text{eff}})$.

Key property (Tier 1, numerically confirmed): s_0 is determined by UV physics (the gap Δ , hopping t , lattice spacing a) and is invariant under all IR variations (system size, subsystem size, boundary conditions) to machine precision. This was confirmed on the free-fermion chain in 1D (central charge $c = 1$ recovered to within 1%; the exact value $c = 1$ is reproduced in a near-critical fitting window, while a full-range fit yields an effective $c_{\text{eff}} \approx 1.4$ due to non-universal lattice corrections far from criticality—see §VII C for details) and on a 2D square-lattice CDW insulator (CV = 0.01% for system size, 0.00% for position).

C. The Emergent Metric

On the one-dimensional envelope of states maximizing entanglement for a given correlation length, the emergent Lorentzian metric takes the form

$$ds^2 = f^2(\ell) dT^2 - d\ell^2, \quad (3)$$

where ℓ is the Bures arc length along the envelope, T is the modular flow parameter, and

$$f^2 \propto \frac{1}{c s_0} \quad (4)$$

is the constitutive lapse-entropy relation.

Framework assumption (constitutive relation). The relation (4) is a framework input, not a derivation. It bridges the quantum information geometry (Bures metric, positive-definite, on parameter space) to the emergent spacetime geometry (Lorentzian, on physical space). Different constitutive relations would produce different field equations. The physical content of this framework is therefore the combination of entanglement equilibrium *plus* this constitutive relation. This is analogous to thermodynamics, where the equation of state (e.g., $PV = nRT$) is an input that encodes the microscopic physics.

¹ Throughout this paper, claims are graded as follows. *Tier 1*: verified mathematics or confirmed numerics. *Tier 2*: framework-internal derivations (structurally necessary consequences of the construction). *Tier 3*: interpretive extensions or conjectures.

1. Signature Emergence

The Bures/Fubini-Study metric is positive-definite (Riemannian). The emergent metric (3) is Lorentzian. The minus sign—the causal structure, the light cones—is not derived from the Bures metric alone.

Lorentzian signature is an input assumption in this framework. The time direction T is identified with modular flow (the one-parameter group generated by $K_{\text{mod}} = -\ln \rho_A$). Modular flow is mathematically well-defined and generates boosts in the emergent geometry (Bisognano-Wichmann theorem [12]), providing a natural candidate for Lorentzian time. However, the identification of modular flow with physical time, and the emergence of the relative sign, is a framework postulate, not a theorem.

The modular flow approach of Liu [13] provides a promising framework for deriving Lorentzian signature from two-sided modular flow on thermofield-double-like structures.

D. The Variational Principle

The variational principle is entanglement equilibrium [3]:

$$\delta S_{\text{ent}}|_V = 0 \quad \text{for all local causal diamonds, at fixed diamond volume.} \quad (5)$$

This condition demands that the entanglement entropy is stationary under small perturbations of the state within any causal diamond—the quantum analog of the Clausius relation $\delta Q = T \delta S$ applied to local Rindler horizons.

III. THE ENTANGLEMENT-GEOMETRY FEEDBACK LOOP

A. The Loop

The self-consistency structure has three legs:

1. **Entanglement $\rightarrow$ Geometry:** The Bures metric and area-law coefficient s_0 determine the emergent metric.
2. **Geometry $\rightarrow$ Causal Structure:** The emergent metric defines light cones, causal diamonds, and domains of dependence.
3. **Causal Structure $\rightarrow$ Entanglement Constraint:** The entanglement entropy of a region can only change consistently with the area theorem and the null Raychaudhuri equation.

B. The Self-Consistency Condition

The geometry created by the entanglement must be such that the entanglement, evolved on that geometry,

reproduces the geometry. The mathematical expression of this condition is the first law of entanglement:

$$\delta S_{\text{ent}} = \delta \langle K_{\text{mod}} \rangle, \quad (6)$$

where K_{mod} is the modular Hamiltonian. When K_{mod} is identified with the geometric modular Hamiltonian (the boost generator on the causal diamond), and the area variation is expanded to second order in the diamond's proper radius ε , the self-consistency condition becomes a local field equation.

The feedback loop (§III A) is the *motivation* for why the self-consistency condition should hold. The *mathematical engine* that produces the field equation is Jacobson's entanglement equilibrium [3]: $\delta S_{\text{ent}} = \delta \langle K_{\text{mod}} \rangle$ applied to small causal diamonds at fixed volume.

What is genuinely new in our construction is not the variational principle itself but three consequences specific to our approach: (1) the emergence of s_0 as a dynamical dilaton field (Jacobson treats the area-law coefficient as a constant related to G_N), (2) the constitutive lapse-entropy relation (4) which connects the Bures metric to the emergent Lorentzian geometry, and (3) the three-way genre classification that follows from considering different entanglement scaling laws—a structure absent from Jacobson's framework.

In 2D specifically, Jacobson's entanglement equilibrium yields the identity $G_{\mu\nu} = 0$ (trivial in 2D) and provides no dynamical content. Our construction, by treating s_0 as a dynamical field, produces the non-trivial entanglement-dilaton gravity equation (7).

IV. THE 2D CASE: ENTANGLEMENT-DILATON GRAVITY

A. The Field Equation

In 1+1 dimensions, the Einstein tensor vanishes identically. The self-consistency condition from §III B, applied to the emergent metric (3), yields:

$$R = -\frac{2}{s_0} \square s_0, \quad (7)$$

where $R = 2f''/f$ is the 2D Ricci scalar and $\square$ is the covariant d'Alembertian. This is a 2D entanglement-dilaton gravity equation with dilaton $\Phi = s_0$.

The full derivation is provided in Appendix A, following the chain from Jacobson's entanglement equilibrium through the CFT₂ modular Hamiltonian [12, 14] and the Callebaut-Verlinde construction [8].

Status of the field equation. Eq. (7) is *exact*, not an approximation: it is the trace of the full tensor equation (9) (contract with $g^{\mu\nu}$ in $D = 2$, where $g^{\mu\nu}g_{\mu\nu} = 2$):

$$\square s_0 - 2\square s_0 - \frac{1}{2} s_0 R = 0 \implies R = -\frac{2\square s_0}{s_0}. \quad (8)$$

TABLE I. Self-consistent 2D solutions.

Parameters	Geometry	Physical interpretation
$a = 0$	$s_0 = \text{const}, R = 0$	Flat space
$b = 0$	$s_0 = a^2 \ell^2, R = -8/(9\ell^2)$	2D dilaton black hole
$a, b > 0$	s_0 grows quadratically	Asymp. flat, horizon at $\ell = -b/a$

The 2D Weyl identity $R = -2\Box \ln s_0$ is a *different* equation (it expands the logarithm: $\Box \ln s_0 = \Box s_0/s_0 - (\nabla s_0)^2/s_0^2$). The two coincide only when $(\nabla s_0)^2 = 0$.

This is not Jackiw-Teitelboim (JT) gravity, which requires $R = \text{const}$. Our curvature $R = -8a^2/[9(a\ell + b)^2]$ is position-dependent.

Tier 2—derived from entanglement equilibrium within the construction.

B. Self-Consistent Solutions

The full tensor equation

$$\nabla_\mu \nabla_\nu s_0 - g_{\mu\nu} \Box s_0 - \frac{1}{4} g_{\mu\nu} s_0 R = 0 \quad (9)$$

(with $\alpha = 1/4$ for $D = 2$; see §V) couples f and s_0 . A natural ansatz that solves the coupled system is the ODE:

$$\ddot{s}_0 = \frac{(\dot{s}_0)^2}{2s_0}, \quad (10)$$

with general solution

$$s_0(\ell) = (a\ell + b)^2. \quad (11)$$

The off-diagonal component of the tensor equation (9) then determines the lapse:

$$3 \frac{f'}{f} s'_0 + s''_0 = 0 \implies f \propto (s'_0)^{-1/3} \propto (a\ell + b)^{-1/3}. \quad (12)$$

The corresponding geometries are shown in Table I.

C. Gauge Stability

Linearizing $s_0 = (a\ell + b)^2 + \epsilon$ about the self-consistent solution gives the Euler equation

$$\epsilon'' - \frac{2a}{x} \epsilon' + \frac{2a^2}{x^2} \epsilon = 0, \quad x = a\ell + b. \quad (13)$$

Solutions: $\epsilon = C_1 x + C_2 x^2$. Both modes are gauge transformations:

- **Mode 1** ($\epsilon \propto x$): coordinate reparametrization $\ell \rightarrow \ell + \epsilon_0$.
- **Mode 2** ($\epsilon \propto x^2$): conformal rescaling $s_0 \rightarrow \lambda s_0$.

In 2D dilaton gravity, there are no propagating gravitational degrees of freedom. All perturbation modes are gauge. The equilibrium is **gauge-stable**.

Caveat: In $D \geq 3$, physical gravitational degrees of freedom exist (gravitational waves, Jeans instability). The 2D gauge-stability result does not extend to higher dimensions without a proper gauge-invariant perturbation analysis.

Tier 2—linearization confirmed independently by two architectures to 10^{-14} precision.

D. The Entanglement-Dilaton Classification Theorem

The most general two-derivative 2D dilaton gravity action is

$$S = \frac{1}{16\pi G_2} \int d^2x \sqrt{-g} [\Phi R + U(\Phi) (\partial\Phi)^2 - 2V(\Phi)]. \quad (14)$$

A well-known classification result [15, 16] states that all models with $U = 0$ (always reachable by field redefinition) form a family parameterized by the arbitrary function $V(\Phi)$.

Theorem 1 (Entanglement-Dilaton Classification). *Given a 1+1 dimensional CFT with central charge c and the universal vacuum entanglement entropy $S = (c/3) \ln(L/\epsilon)$, the entanglement equilibrium condition $\delta S_{\text{ent}} = 0$ on local causal diamonds, together with the constitutive relation (4), selects a unique equivalence class of 2D dilaton gravity theories (up to $V(\Phi)$). Within this class:*

- The dilaton is $\Phi = s_0 \propto c$.
- The field equations are

$$\nabla_\mu \nabla_\nu \Phi - \frac{1}{2} g_{\mu\nu} \Box \Phi = 0, \quad R = -\frac{2}{\Phi} \Box \Phi. \quad (15)$$

- All actions whose solutions satisfy these equations (after possible field redefinitions) belong to the same equivalence class.

The class is parameterized by the asymptotic behavior of $V(\Phi)$, which is not fixed by the CFT data alone but must be supplied by additional input (boundary conditions, matter coupling, or asymptotic data). The physical observables—entropy, geometry, causal structure—are universal within the class; the remaining freedom in $V(\Phi)$ is physical (different V give different solutions) and must be supplied externally.

Corollary. The central charge c enters only as an overall normalization of the dilaton and can be absorbed into the definition of the 2D Planck scale G_2 .

Tier 2—proven within the framework (contingent on Assumption A6, the constitutive relation). The identification $\Phi = s_0$ is a dictionary entry analogous to the

AdS/CFT bulk-boundary dictionary; the uniqueness of the equivalence class under field redefinitions follows from the Grumiller *et al.* classification [15]. *Note:* Earlier versions of this paper labeled this result a Conjecture. The upgrade to Theorem reflects the completion of the derivation chain in Appendix A, which closes the gap that previously made the status uncertain.

V. THE HIGHER-DIMENSIONAL GENERALIZATION (STRUCTURED ANSATZ)

A. The Generalized Field Equation

In $D = d+1$ spacetime dimensions, we propose the following structured ansatz generalizing the 2D self-consistency condition:

$$s_0 G_{\mu\nu} = \nabla_\mu \nabla_\nu s_0 - g_{\mu\nu} \square s_0 - \alpha_D g_{\mu\nu} s_0 R, \quad (16)$$

where $G_{\mu\nu}$ is the Einstein tensor and the dimension-dependent coefficient is

$$\alpha_D = \begin{cases} 1/4 & D = 2 \text{ (from entanglement equilibrium)}, \\ \frac{D-2}{2D} & D > 2 \text{ (from constant-} s_0 \rightarrow \text{Einstein)}. \end{cases} \quad (17)$$

Honest status. This equation is a *structured ansatz*, not a first-principles derivation from D -dimensional entanglement equilibrium. The procedure is: (1) write the most general tensor equation involving $G_{\mu\nu}$, $\nabla_\mu \nabla_\nu s_0$, $g_{\mu\nu} \square s_0$, and $g_{\mu\nu} s_0 R$; (2) impose that constant s_0 reproduces vacuum Einstein; (3) this fixes α_D . The tensor structure corresponds to Brans-Dicke theory with $\omega = 0$, equivalent to $f(R)$ gravity—a well-studied scalar-tensor framework.

A full derivation from D -dimensional entanglement equilibrium with dynamical s_0 remains an open problem (§IX).

Tier 2—structural generalization constrained by consistency with the derived 2D result and the Einstein limit.

B. The Einstein Consistency Check

When $s_0 = \text{const}$, for all $D > 2$, Eq. (16) reduces to

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 0, \quad \Lambda = \alpha_D R, \quad (18)$$

which is vacuum Einstein gravity with a cosmological constant in **every dimension** $D > 2$. The cosmological constant Λ can be positive (de Sitter), negative (anti-de Sitter), or zero (Ricci-flat), depending on boundary conditions.

C. The 2D Limit

Setting $D = 2$: $G_{\mu\nu} \equiv 0$ and $\alpha_2 = 1/4$. Equation (16) reduces to (9), whose trace gives $R = -(2/s_0) \square s_0$ —

recovering the established 2D entanglement-dilaton gravity result.

D. Jacobson Agreement

Jacobson [4] showed $\delta Q = T \delta S$ on local Rindler horizons implies $G_{\mu\nu} = 8\pi G T_{\mu\nu}$. His 2016 entanglement equilibrium argument [3] yields the linearized Einstein equation from $\delta S_{\text{ent}} = \delta \langle K_{\text{mod}} \rangle$ for conformal fields. Our construction uses the same variational principle, so agreement in the constant- s_0 limit is expected, not discovered—both constructions rest on $\delta S_{\text{ent}} = \delta \langle K_{\text{mod}} \rangle$.

What is genuinely distinct from Jacobson: (1) treating s_0 as a dynamical field produces scalar-tensor gravity (Jacobson’s construction yields pure Einstein with fixed G_N); (2) the constitutive relation (4) provides an explicit bridge from quantum state data to the emergent metric; (3) the genre-locking classification extends the framework to non-area-law entanglement.

VI. GENRE-LOCKING: THE CLASSIFICATION

A. The Hypothesis

Different entanglement scaling laws naturally select different gravitational field equations as their self-consistency conditions. The specific gravity is not imposed from outside but emerges as the unique self-consistent equilibrium for a given entanglement “genre.”

B. Distinction from Functional-Form Selection

Prior work [1, 2, 7] established that the *functional form* of entanglement entropy selects the gravitational theory: Ryu-Takayanagi (area) $\rightarrow$ Einstein; Wald functional $\rightarrow$ higher-curvature gravity. This operates *within* the area-law regime.

Genre-locking operates *across* scaling regimes: area-law vs. log-law vs. volume-law. These are qualitatively different entanglement structures, not different functionals within one regime. The selection principle is orthogonal to functional-form selection and classifies a broader landscape.

C. The Phase Diagram

The volume-law arm is the least developed. In random matrix theory and thermal states above the Page temperature, entanglement is maximal and structureless—precisely the regime where no coherent geometric description is expected. Physically, volume-law scaling means every degree of freedom is maximally entangled with every other; there is no preferred bipartition, hence no boundary, hence no area term to anchor an emergent

TABLE II. Genre-locking phase diagram. The “Novel contribution” column identifies what is new relative to the cited prior art.

Genre	Scaling	Emergent gravity	Prior art	Novel contribution
Area-law (gapped)	$S \sim s_0 \cdot A(\partial R)$	Einstein / scalar-tensor ($D > 2$); dilaton (2D)	[1, 3]	Self-consistency via Bures metric; s_0 as dilaton
Log-law (critical, CFT)	$S \sim (c/3) \ln L$	2D dilaton ($R \neq 0$ in general; $R \rightarrow 0$ at criticality as $s_0 \rightarrow \infty$)	[8]	Classification within phase diagram
Volume-law (thermal)	$S \sim \text{Volume}$	No coherent semiclassical metric	[17]	Least-explored arm; novelty whitespace

metric via Eq. (2). The Bures metric on such states degenerates: all directions in parameter space are equally “far,” yielding a featureless geometry with no light-cone structure. Li, Chen, and Fisher [17] demonstrated a measurement-induced entanglement transition between area-law and volume-law phases; the volume-law phase in their work exhibits no coherent geometric structure, consistent with the genre-locking prediction. This aligns with the general expectation from holographic error-correction codes [18]: states saturating the Page bound lack the redundancy structure needed to define a bulk geometry. Developing this arm further—connecting to scrambling dynamics, quantum error-correction thresholds, and non-local holographic entanglement entropy results—is an important direction for future work.

D. The Phase Transition Prediction

At a quantum phase transition (gapped $\rightarrow$ critical), the area-law coefficient $s_0 \rightarrow \infty$ and the curvature $R \rightarrow 0$. **Gravity “turns off” at criticality.** This is a falsifiable prediction testable on quantum lattice systems.

E. Uniqueness

For a given entanglement genre, the self-consistency condition selects a unique equivalence class of gravitational theories (see Theorem 1 for the precise 2D statement). The genre $\rightarrow$ gravity mapping is unique at the level of physics.

VII. THE CIRCULARITY RESOLUTION

A. The $N = 2$ Trap

On the bare two-qubit CHSH envelope, the lapse normalization is underivable in principle. Jacobson’s construction needs an area term to balance δS_{ent} , and two qubits have no spatial boundary—no area. The “independent input” (Bures volume) turned out to be determined by the same Schmidt spectrum, making the derivation circular.

B. The Many-Body Resolution

In the many-body gapped regime, an area law emerges with coefficient s_0 playing the role of $1/(4G_N)$. The crowd supplies the structure missing at $N = 2$.

C. Numerical Confirmation (Tier 1)

1D free-fermion chain ($L = 2000$, $n_{\text{sub}} = 800$, periodic boundary conditions, Peschel correlation-matrix method [19]): The central charge $c = 1$ is recovered to within 1% in a near-critical fitting window ($\xi/a > 10$). A full-range fit (all $\Delta/t \in [0.01, 5.0]$) yields $c_{\text{eff}} \approx 1.4$ due to non-universal corrections far from criticality, where the logarithmic scaling $S \sim (c/3) \ln(L/\varepsilon)$ no longer holds. The distinction illustrates the importance of specifying the fitting window when extracting universal data from finite-size numerics. s_0 is invariant under all IR variations (system size L , subsystem size n , boundary conditions) at fixed UV parameters (Δ, t, a).

2D square-lattice CDW insulator: s_0 scales linearly with perimeter (area law in 2D) with clean extraction across $\Delta/t \in [0.1, 5.0]$. Independence check: system-size CV = 0.01%, position CV = 0.00%, shape CV = 0.08% (corner corrections). Filling-fraction change (a UV modification) correctly produces a change in s_0 , confirming it responds to UV physics as predicted.

VIII. FALSIFIABLE PREDICTIONS

- 1. Phase transition gravity shutdown:** At the quantum critical point (gapped $\rightarrow$ critical), $R \rightarrow 0$ and gravity effectively turns off. Testable on quantum lattice systems.
- 2. Logarithmic scaling of s_0 near criticality:** From the Calabrese-Cardy formula [20], the area-law coefficient in the gapped regime scales as

$$s_0 = \frac{c}{6} \ln\left(\frac{\xi}{a}\right) + \text{const}, \quad (19)$$

where $\xi \sim |g - g_c|^{-\nu}$ is the correlation length. This yields exact, parameter-free slopes: $c/6 = 1/12$

for the transverse-field Ising model ($c = 1/2$) and $c/6 = 1/6$ for the CDW free-fermion chain ($c = 1$). The emergent curvature therefore vanishes *logarithmically* near criticality,

$$R \sim \frac{1}{\ln|g - g_c|^{-1}} \rightarrow 0, \quad (20)$$

not as a power law. This is directly testable: plot s_0 vs. $\ln \xi$ and verify a straight line with the exact Calabrese-Cardy slope. *Verification recipe:* compute $s_0(\ell)$ from the entanglement entropy of contiguous blocks, fit to $(a\ell + b)^2$, extract $R = -8a^2/[9(a\ell + b)^2]$, and verify the scaling collapse as $g \rightarrow g_c$.

3. **Genre-gravity correspondence:** Different entanglement scaling laws produce different self-consistency conditions. Volume-law states should produce no coherent semiclassical emergent metric.
4. **Scalar-tensor signatures:** If s_0 varies in a physical system, the emergent gravity should deviate from Einstein's equation in the specific Brans-Dicke pattern predicted by Eq. (16), with s_0 as the scalar field.
5. **2D solutions:** The self-consistent entanglement profiles $s_0(\ell) = (a\ell + b)^2$ should match 2D entanglement-dilaton black hole geometries.

IX. OPEN QUESTIONS

1. **$D \geq 3$ stability:** The gauge-stability result in 2D does not extend to higher dimensions where physical gravitational degrees of freedom exist. Gravitational instability (Jeans) is expected and would be physically correct.
2. **Coefficient verification:** $\alpha_D = (D-2)/(2D)$ for $D > 2$ must be verified against Jacobson's explicit calculations for each D .
3. **Matter sector:** The identification $T_{\mu\nu}$ = deviations from the maximal-entropy envelope is structural; the detailed mapping remains open.
4. **Cosmological constant origin:** Λ is a free parameter in all $D > 2$. Whether its value connects to the entanglement structure at a deeper level is unknown.
5. **Volume-law arm:** The volume-law $\rightarrow$ no-coherent-geometry arm is the least explored.
6. **Formal vs. dynamical time:** The emergent lapse $f^2 \propto 1/\ln(\xi)$ is inverse-logarithmic while physical observables are power-law in ξ . This may indicate the emergent "time" is information-geometric rather than dynamical.

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Appendix A: Derivation of the 2D Entanglement-Dilaton Field Equation

From Entanglement Equilibrium to $R = -2\Box s_0/s_0$ — Step by Step.

1. Setup and Assumptions

We state the independent assumptions explicitly:

A1 (Entanglement Equilibrium [3]). For a small causal diamond $D(\Sigma)$ of proper radius ε in the emergent geometry, the entanglement entropy of the vacuum is stationary under first-order variations at fixed diamond volume: $\delta S_{\text{ent}}|_V = 0$. *Status: Framework postulate.*

A2 (Area-law UV structure). The entanglement entropy admits a UV/IR decomposition: $S_{\text{ent}} = s_0 \cdot A(\partial R) + S_{\text{IR}}$. In 2D, $A(\partial R)$ = number of boundary points (typically 2 for an interval). *Status: Empirical input (Tier 1).*

A3 (Dynamical s_0). The area-law coefficient s_0 is promoted from a constant to a field $s_0(x)$. This is the key modeling choice that makes the 2D construction non-trivial and distinguishes it from Jacobson's framework. *Status: Modeling assumption, new in this work.*

A4 (Local modular Hamiltonian [12, 14]). For a CFT vacuum reduced to a causal diamond, the modular Hamiltonian is a local integral of the stress tensor: $K = 2\pi \int_{\Sigma} \zeta^a T_{ab} d\Sigma^b$, where ζ is the conformal Killing vector generating the diamond's causal flow. *Status: Theorem for CFT vacua on ball/diamond regions.*

A5 (Lorentzian signature). The emergent metric is Lorentzian: $ds^2 = f^2(\ell) dT^2 - d\ell^2$. *Status: Input (see §II C 1).*

A6 (Constitutive relation). $f^2 \propto 1/(c s_0)$. *Status: Framework input, new in this work.*

2. The Jacobson Chain ($D \geq 3$, constant s_0)

Before specializing to 2D, we reproduce Jacobson’s argument.

Step 1: Entropy variation decomposition. With constant s_0 ($= \eta$ in Jacobson’s notation):

$$\delta S_{\text{ent}} = \eta \cdot \delta A + \delta S_{\text{IR}}. \quad (\text{A1})$$

Step 2: Entanglement first law. The vacuum reduced to the diamond is formally thermal with $T = \hbar/(2\pi)$. Free-energy stationarity gives:

$$\delta S_{\text{IR}} = \frac{2\pi}{\hbar} \delta \langle K \rangle. \quad (\text{A2})$$

Step 3: Local modular Hamiltonian (A4). For slowly varying excitations:

$$\delta \langle H_\zeta \rangle = \frac{\Omega_{d-2} \ell^d}{d^2 - 1} \delta \langle T_{00} \rangle. \quad (\text{A3})$$

Step 4: Geometric area deficit at fixed volume.

$$\delta A|_V = -\frac{\Omega_{d-2} \ell^d}{d^2 - 1} G_{00}. \quad (\text{A4})$$

Step 5: Combine $\rightarrow$ Einstein equation. The geometric prefactors cancel:

$$G_{ab} + \Lambda g_{ab} = \frac{2\pi}{\hbar\eta} \langle T_{ab} \rangle = 8\pi G \langle T_{ab} \rangle, \quad (\text{A5})$$

with $G = 1/(4\hbar\eta)$.

3. The 2D Obstruction

In $D = 2$, $G_{\mu\nu} \equiv 0$ identically. Step 4 gives $\delta A|_V \propto G_{00} = 0$, and the equilibrium condition constrains the matter but produces no gravitational dynamics. The resolution: invoke Assumption A3—promote s_0 to a dynamical field.

4. The 2D Derivation with Dynamical s_0

Step 1: Modified entropy decomposition.

$$\delta S_{\text{ent}} = (\delta s_0) \cdot A + s_0 \cdot (\delta A) + \delta S_{\text{IR}}. \quad (\text{A6})$$

The first term—absent in Jacobson’s constant- η argument—sources all 2D dilaton dynamics.

Step 2: Identify leading entropy with the metric (Callebaut-Verlinde [8]). In a 2D CFT with central

charge c , the vacuum entanglement entropy of an interval $[u, v]$ is the Calabrese-Cardy/Holzhey-Larsen-Wilczek log-law [20, 21]. Following Callebaut and Verlinde, the rescaled entropy is identified with the conformal mode of a constant-curvature 2D metric: $ds^2 = -e^{-\tilde{S}_0} du dv$, with $R_0 = -2/\ell^2$. The leading entropy *is* the metric.

Step 3: Identify subleading entropy with the dilaton. Decompose $S = S_0 + S_1$, where $S_1 = \langle K \rangle$. The entanglement first law $\delta S_1 = \delta \langle K \rangle$ yields:

$$\nabla_+ \partial_+ S_1 = 2\pi \langle T_{++} \rangle, \quad \nabla_- \partial_- S_1 = 2\pi \langle T_{--} \rangle. \quad (\text{A7})$$

Callebaut and Verlinde identify $\frac{1}{4}\Phi = -S_1 = -\langle K \rangle$ and show these are “identical to the equations of motion of the dilaton in JT gravity.”

Step 4: Extract the field equation. The trace of the tensor equation (9) (contracting with $g^{\mu\nu}$ in $D = 2$) gives exactly $R = -(2/s_0)\square s_0$, which is Eq. (7). This is exact—not an approximation—as shown in §IV A.

5. Summary: Assumptions vs. Derivations

TABLE III. Status of each step in the derivation chain.

Step	Content	Status
A1	Entanglement equilibrium	Assumed
A2	Area-law UV structure	Empirical (Tier 1)
A3	s_0 promoted to field	Assumed (new)
A4	$K_{\text{mod}} = \int \zeta \cdot T$	Theorem (BW/CHM)
A5	Lorentzian signature	Assumed
A6	$f^2 \propto 1/(cs_0)$	Assumed (new)
$D \geq 3$ chain	Einstein equation	Derived [3]
2D: $S_0 = \text{metric}$	Conformal factor	Derived [8]
2D: $S_1 = \text{dilaton}$	$\langle K \rangle = \text{dilaton}$	Derived [8]
2D: Dilaton EOM	$\nabla \partial S_1 = 2\pi T$	Derived
2D: $R = -2\square s_0/s_0$	Weyl identity + approx	Derived (exact: $R = -2\square$)

The two genuinely new inputs (not present in Jacobson or Callebaut-Verlinde) are Assumptions A3 and A6.

6. Connection to the Higher-Dimensional Generalization

The higher-dimensional field equation (16) is *not* derived from entanglement equilibrium in $D > 2$. It is a structured ansatz constrained by requiring: (1) correct 2D limit ($\alpha_2 = 1/4$); (2) vacuum Einstein when $s_0 = \text{const}$; (3) the tensor structure of Brans-Dicke theory with $\omega = 0$. A first-principles derivation from D -dimensional entanglement equilibrium with dynamical s_0 remains an open problem.

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